

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 4.5 — Managing Networking Resources

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Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 4: Ensuring successful operation of a cloud solution (~20 % of the exam)**. You must know how to add and expand subnets, reserve static IPs, and work with Cloud DNS and Cloud NAT.

1. Adding a Subnet to an Existing VPC

Creating a New Subnet

```
# Add a subnet to an existing custom VPC
gcloud compute networks subnets create new-subnet \
  --network=my-vpc \
  --region=europe-west1 \
  --range=10.1.0.0/24

# Add a subnet with Private Google Access
gcloud compute networks subnets create private-subnet \
  --network=my-vpc \
  --region=asia-east1 \
  --range=10.2.0.0/24 \
  --enable-private-ip-google-access

# Add a subnet with VPC Flow Logs
gcloud compute networks subnets create logged-subnet \
  --network=my-vpc \
  --region=us-west1 \
  --range=10.3.0.0/24 \
  --enable-flow-logs \
  --logging-aggregation-interval=interval-5-sec \
  --logging-flow-sampling=0.5

# Add a subnet with secondary ranges for GKE
gcloud compute networks subnets create gke-subnet \
  --network=my-vpc \
  --region=us-central1 \
  --range=10.4.0.0/24 \
  --secondary-range=pod-range=10.8.0.0/14,service-range=10.12.0.0/20
```

Key Considerations When Adding Subnets

- Subnet IP ranges **cannot overlap** with other subnets in the same VPC
 - Subnets are **regional** — they span all zones in a region
 - VPC must be in **custom mode** (auto mode creates subnets automatically)
 - You can add subnets without downtime or affecting existing resources
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2. Expanding a Subnet to Have More IP Addresses

Expanding Subnet CIDR Range

You can **expand** a subnet’s primary IP range without downtime. This is useful when you’re running out of IP addresses.

```
# Expand a subnet (increase the range)
gcloud compute networks subnets expand-ip-range my-subnet \
  --region=us-central1 \
  --prefix-length=20

# Example: Expand from /24 (256 IPs) to /20 (4,096 IPs)
# Original: 10.0.1.0/24 → Expanded: 10.0.0.0/20
```

Rules for Subnet Expansion

- You can only **expand** (make larger), never shrink
- The new range must **contain the original range**
- The operation is **non-disruptive** (no downtime)
- Cannot overlap with other subnets
- The expanded range must be a valid CIDR block

CIDR Reference

CIDR	Subnet Mask	Usable IPs	Example
/28	255.255.255.240	12	Small test environments
/24	255.255.255.0	252	Standard subnet
/20	255.255.240.0	4,092	Medium workloads
/16	255.255.0.0	65,532	Large workloads
/12	255.240.0.0	1,048,572	Very large workloads

Note: Google reserves 4 IPs per subnet (network, gateway, reserved, broadcast), so usable IPs = total - 4.

3. Reserving Static IP Addresses

Types of IP Addresses

Type	Scope	Use Case
Ephemeral external	Releases when VM stops	Dev/test VMs
Static external	Persists independently	Production VMs, DNS, load balancers
Ephemeral internal	Auto-assigned from subnet range	Standard VMs
Static internal	Specific internal IP reserved	Databases, DNS servers

Reserving External Static IPs

```
# Reserve a regional external IP
gcloud compute addresses create my-external-ip \
  --region=us-central1

# Reserve a global external IP (for global load balancers)
gcloud compute addresses create my-global-ip \
  --global

# Reserve with a specific network tier
gcloud compute addresses create standard-ip \
  --region=us-central1 \
  --network-tier=STANDARD

# List reserved addresses
gcloud compute addresses list

# Describe a reserved address
gcloud compute addresses describe my-external-ip --region=us-central1

# Get just the IP address
gcloud compute addresses describe my-external-ip \
  --region=us-central1 \
  --format="get(address)"
```

Assigning a Static External IP to a VM

```
# At creation time
gcloud compute instances create my-vm \
  --zone=us-central1-a \
  --address=my-external-ip

# To an existing VM (must delete and recreate the access config)
gcloud compute instances delete-access-config my-vm \
  --zone=us-central1-a \
  --access-config-name="External NAT"

gcloud compute instances add-access-config my-vm \
  --zone=us-central1-a \
  --address=MY_STATIC_IP
```

Reserving Internal Static IPs

```
# Reserve a specific internal IP from a subnet
gcloud compute addresses create my-internal-ip \
  --region=us-central1 \
  --subnet=my-subnet \
  --addresses=10.0.1.50

# Reserve any available internal IP from a subnet
gcloud compute addresses create auto-internal-ip \
  --region=us-central1 \
  --subnet=my-subnet

# Use the reserved internal IP when creating a VM
gcloud compute instances create my-vm \
  --zone=us-central1-a \
  --network-interface=network=my-vpc,subnet=my-subnet,private-network-ip=10.0.1.50,no-address
```

Releasing Static IPs

```
# Release (delete) a reserved IP
gcloud compute addresses delete my-external-ip --region=us-central1

# Release a global IP
gcloud compute addresses delete my-global-ip --global
```

Important: Unused reserved static IPs incur charges. Release them when no longer needed.

4. Working with Cloud DNS

What Is Cloud DNS?

- **Managed authoritative DNS** service
- 100% uptime SLA for authoritative name serving
- Supports public and private zones
- Supports DNSSEC

DNS Zone Types

Type	Visibility	Use Case
Public	Internet	External-facing services
Private	VPC networks only	Internal services
Forwarding	VPC → on-premises DNS	Hybrid cloud
Peering	VPC → another VPC's DNS	Cross-project DNS

Managing Public DNS Zones

```
# Create a public DNS zone
gcloud dns managed-zones create my-zone \
  --dns-name=example.com. \
  --description="My public DNS zone" \
  --visibility=public

# List zones
gcloud dns managed-zones list

# Describe a zone
gcloud dns managed-zones describe my-zone
```

Managing DNS Records

```
# Start a DNS transaction
gcloud dns record-sets transaction start --zone=my-zone

# Add an A record
gcloud dns record-sets transaction add "203.0.113.1" \
  --name=www.example.com. \
  --ttl=300 \
  --type=A \
  --zone=my-zone

# Add a CNAME record
gcloud dns record-sets transaction add "example.com." \
  --name=blog.example.com. \
  --ttl=300 \
  --type=CNAME \
  --zone=my-zone

# Add an MX record
gcloud dns record-sets transaction add "10 mail.example.com." \
  --name=example.com. \
  --ttl=300 \
  --type=MX \
  --zone=my-zone

# Execute the transaction
gcloud dns record-sets transaction execute --zone=my-zone

# Abort a transaction
gcloud dns record-sets transaction abort --zone=my-zone

# List records in a zone
gcloud dns record-sets list --zone=my-zone

# Delete a record (via transaction)
gcloud dns record-sets transaction start --zone=my-zone
gcloud dns record-sets transaction remove "203.0.113.1" \
  --name=www.example.com. \
  --ttl=300 \
  --type=A \
  --zone=my-zone
gcloud dns record-sets transaction execute --zone=my-zone
```

Managing Private DNS Zones

```
# Create a private DNS zone (visible only within specified VPCs)
gcloud dns managed-zones create my-private-zone \
  --dns-name=internal.example.com. \
  --description="Internal DNS" \
  --visibility=private \
  --networks=my-vpc

# Add the zone to additional VPCs
gcloud dns managed-zones update my-private-zone \
  --networks=my-vpc,my-other-vpc
```

DNS Forwarding (Hybrid Cloud)

```
# Create a forwarding zone (forward queries to on-premises DNS)
gcloud dns managed-zones create on-prem-zone \
  --dns-name=corp.example.com. \
  --description="Forward to on-prem DNS" \
  --visibility=private \
  --networks=my-vpc \
  --forwarding-targets=192.168.1.10,192.168.1.11
```


Common DNS Record Types

Type	Description	Example
A	IPv4 address	<code>www.example.com → 203.0.113.1</code>
AAAA	IPv6 address	<code>www.example.com → 2001:db8::1</code>
CNAME	Canonical name (alias)	<code>blog.example.com → example.com</code>
MX	Mail exchange	<code>example.com → 10 mail.example.com</code>
TXT	Text record	SPF, DKIM, domain verification
NS	Name server	Zone delegation
SOA	Start of authority	Zone metadata
SRV	Service location	Service discovery
PTR	Reverse DNS	IP → hostname

5. Working with Cloud NAT

What Is Cloud NAT?

- Provides **outbound internet access** for VMs without external IPs
- Software-defined, fully managed (no NAT VMs or appliances)
- Works with Compute Engine and GKE
- **Regional resource** associated with a Cloud Router

Setting Up Cloud NAT

```
# Step 1: Create a Cloud Router (if not already created)
gcloud compute routers create my-router \
  --network=my-vpc \
  --region=us-central1

# Step 2: Create Cloud NAT configuration
gcloud compute routers nats create my-nat \
  --router=my-router \
  --region=us-central1 \
  --auto-allocate-nat-external-ips \
  --nat-all-subnet-ip-ranges

# Create NAT for specific subnets only
gcloud compute routers nats create my-nat \
  --router=my-router \
  --region=us-central1 \
  --auto-allocate-nat-external-ips \
  --nat-custom-subnet-ip-ranges=my-subnet

# Create NAT with manual IP allocation
gcloud compute routers nats create my-nat \
  --router=my-router \
  --region=us-central1 \
  --nat-external-ip-pool=my-nat-ip-1,my-nat-ip-2 \
  --nat-all-subnet-ip-ranges
```

Managing Cloud NAT

```
# List NAT configurations
gcloud compute routers nats list --router=my-router --region=us-central1

# Describe a NAT configuration
gcloud compute routers nats describe my-nat \
  --router=my-router \
  --region=us-central1

# Update NAT configuration
gcloud compute routers nats update my-nat \
  --router=my-router \
  --region=us-central1 \
  --min-ports-per-vm=128 \
  --max-ports-per-vm=4096

# Enable logging
gcloud compute routers nats update my-nat \
  --router=my-router \
  --region=us-central1 \
  --enable-logging \
  --log-filter=ALL

# Delete NAT configuration
gcloud compute routers nats delete my-nat \
  --router=my-router \
  --region=us-central1
```

Cloud NAT Key Properties

- **Outbound only** — External traffic cannot initiate connections to NAT'd VMs
- **No proxy** — Traffic is not proxied; source IP is rewritten at the network layer
- **Per-region** — Each region needs its own Cloud NAT configuration
- **Port allocation** — Controls how many outbound connections each VM can make
- **Logging** — Can log NAT translations for monitoring and troubleshooting

Cloud NAT vs. External IP

Feature	Cloud NAT	External IP on VM
Outbound internet	Yes	Yes
Inbound from internet	No	Yes
IP management	Centralized	Per-VM
Security	Higher (no public exposure)	Lower (publicly addressable)
Cost	NAT gateway charges	IP address charges
Use case	VMs that need internet but not exposure	Public-facing servers

8. Firewall Rules Management

```
# Create ingress rule
gcloud compute firewall-rules create allow-http \
  --network=MY_VPC \
  --direction=INGRESS \
  --priority=1000 \
  --action=ALLOW \
  --rules=tcp:80,tcp:443 \
  --source-ranges=0.0.0.0/0 \
  --target-tags=web-server

# Create egress deny rule
gcloud compute firewall-rules create deny-all-egress \
  --network=MY_VPC \
  --direction=EGRESS \
  --priority=65534 \
  --action=DENY \
  --rules=all

# Update existing rule
gcloud compute firewall-rules update allow-http --priority=900

# List rules sorted by priority
gcloud compute firewall-rules list --sort-by=priority --filter="network=MY_VPC"

# Delete rule
gcloud compute firewall-rules delete allow-http
```

Key concepts: - Implied rules cannot be deleted: default-deny-all-ingress (priority 65535) and default-allow-all-egress (priority 65535) - Lower priority number = evaluated first - DENY at same priority beats ALLOW - Target service account (instead of tag): more secure, prevents tag spoofing `bash --target-service-accounts=SA@PROJECT.iam.gserviceaccount.com --source-service-accounts=SA@PROJECT.iam.gserviceaccount.com`

9. VPC Peering Management

```
# Both sides must create peering connections
gcloud compute networks peerings create peer-a-to-b \
  --network=VPC_A \
  --peer-project=PROJECT_B \
  --peer-network=VPC_B \
  --export-custom-routes \
  --import-custom-routes

# On project B side:
gcloud compute networks peerings create peer-b-to-a \
  --network=VPC_B \
  --peer-project=PROJECT_A \
  --peer-network=VPC_A

# List, describe, delete
gcloud compute networks peerings list --network=VPC_A
gcloud compute networks peerings delete peer-a-to-b --network=VPC_A
```

- Peering is ACTIVE only when both sides are created
 - No transitive routing
 - Overlapping CIDR ranges cannot be peered
 - Exam tip: each VPC can have up to 25 peering connections
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10. Hybrid Connectivity

Cloud VPN (HA VPN)

```
# Create HA VPN gateway (2 interfaces for 99.99% SLA)
gcloud compute vpn-gateways create my-vpn-gw --network=MY_VPC --region=us-central1

# Create external VPN gateway (peer side)
gcloud compute external-vpn-gateways create peer-vpn-gw \
  --interfaces 0=PEER_IP_0,1=PEER_IP_1

# Create VPN tunnels (one per interface pair)
gcloud compute vpn-tunnels create tunnel-1 \
  --peer-external-gateway=peer-vpn-gw \
  --peer-external-gateway-interface=0 \
  --vpn-gateway=my-vpn-gw --vpn-gateway-region=us-central1 \
  --interface=0 --shared-secret=SECRET \
  --router=MY_ROUTER --region=us-central1

# Configure BGP session on Cloud Router
gcloud compute routers add-bgp-peer MY_ROUTER \
  --peer-name=bgp-peer-1 --peer-ip=169.254.0.2 \
  --peer-asn=65001 --interface=vpn-tunnel-1 --region=us-central1
```

Cloud Interconnect Comparison

	Dedicated Interconnect	Partner Interconnect
Bandwidth	10 Gbps or 100 Gbps per link	50 Mbps – 50 Gbps
Setup	Direct peering with Google PoP	Via service provider
SLA	99.99% (4 connections)	99.99% (redundant)
Use when	>10 Gbps or strict latency needs	Can't reach Google PoP directly

Exam tip: Cloud VPN max bandwidth ~3 Gbps per tunnel; for >3 Gbps use Interconnect

Exam Practice Questions

1. You are running out of IP addresses in a /24 subnet. How can you increase the range without downtime?

- Answer: Use `gcloud compute networks subnets expand-ip-range` to expand the prefix length (e.g., from /24 to /20). This is non-disruptive.
2. **You need a stable IP address for a production VM that persists even if the VM is stopped. What should you do?**
- Answer: **Reserve a static external IP** address and assign it to the VM. Static IPs persist independently of the VM lifecycle.
3. **Internal microservices need to resolve custom DNS names like `api.internal.example.com`. How should you configure this?**
- Answer: Create a **private DNS zone** in Cloud DNS with `--dns-name=internal.example.com.` and associate it with the VPC. Add A records for each service.
4. **VMs in a private subnet (no external IPs) need to download packages from the internet. What's the best solution?**
- Answer: Set up **Cloud NAT** on the VPC with a Cloud Router. This provides outbound internet access without exposing VMs to inbound traffic.
5. **You have unused reserved static IP addresses. Are you being charged for them?**
- Answer: **Yes**. Google Cloud charges for reserved static IP addresses that are not assigned to a resource. Release unused IPs to avoid charges.
6. **You need to forward DNS queries for `corp.company.com` to on-premises DNS servers at 10.0.0.53 and 10.0.0.54. How?**
- Answer: Create a **Cloud DNS forwarding zone** with `--dns-name=corp.company.com.` and `--forwarding-targets=10.0.0.53,10.0.0.54`, associated with the appropriate VPC.
7. **Can you shrink a subnet's IP range after expanding it?**
- Answer: **No**. Subnet expansion is a one-way operation. You can only increase the range, never decrease it.

Glossary

A record: A DNS resource record type that maps a hostname to an IPv4 address (e.g., `www.example.com → 203.0.113.1`).

AAAA record: A DNS resource record type that maps a hostname to an IPv6 address.

ASN (Autonomous System Number): A globally unique identifier assigned to a network (or group of networks) that share a routing policy, used in BGP peering configurations for Cloud VPN and Cloud Interconnect.

Auto mode VPC: A VPC network mode in which Google automatically creates one subnet per region with predefined IP ranges; contrasted with custom mode VPCs.

BGP (Border Gateway Protocol): A dynamic routing protocol used by Cloud Router and Cloud VPN tunnels to exchange routes between the Google Cloud network and on-premises or peer networks.

CIDR (Classless Inter-Domain Routing): A compact notation for specifying IP address ranges (e.g., `10.0.0.0/24`), where the suffix indicates the number of fixed bits in the network prefix.

Cloud DNS: Google Cloud's fully managed, authoritative DNS service that provides 100% uptime SLA for name serving and supports public zones, private zones, forwarding zones, and DNSSEC.

Cloud Interconnect: A Google Cloud service that provides direct, high-bandwidth, low-latency physical connectivity between an on-premises network and Google's network, available as Dedicated or Partner Interconnect.

Cloud NAT (Network Address Translation): A fully managed, software-defined Google Cloud service that provides outbound internet access for VMs and GKE nodes that have no external IP addresses, without exposing them to inbound connections.

Cloud Router: A fully managed Google Cloud service that dynamically exchanges routes between a VPC network and on-premises networks using BGP; required by Cloud VPN (HA VPN) and Cloud NAT.

Cloud VPN: A Google Cloud service that provides encrypted connectivity between a VPC network and an on-premises or another cloud network over the public internet using IPsec tunnels; HA VPN offers 99.99% SLA.

CNAME record: A DNS resource record type that maps an alias hostname to the canonical (true) hostname of another domain.

Custom mode VPC: A VPC network mode where subnets must be created manually with user-defined IP ranges, giving full control over subnet regions and address spaces.

Dedicated Interconnect: A Cloud Interconnect option that provides direct physical connections of 10 Gbps or 100 Gbps per link between an on-premises data center and a Google Point of Presence (PoP).

DKIM (DomainKeys Identified Mail): An email authentication method used to sign outgoing email; the public key is published as a TXT DNS record.

DNS (Domain Name System): The hierarchical naming system that translates human-readable domain names (e.g., `example.com`) into IP addresses.

DNSSEC (DNS Security Extensions): A set of extensions that add cryptographic signatures to DNS records, allowing resolvers to verify that DNS responses have not been tampered with.

DNS forwarding zone: A Cloud DNS zone type that forwards DNS queries for a given domain to specified external DNS server IP addresses, used for hybrid cloud DNS resolution.

DNS peering zone: A Cloud DNS zone type that allows one VPC to use DNS zones from another VPC, enabling cross-project DNS resolution.

DNS transaction: A Cloud DNS mechanism for atomically staging and applying multiple record changes (add, remove) to a managed zone using `transaction start`, `add/remove`, and `execute` commands.

Egress: Network traffic that leaves a VPC network or Google's network; firewall egress rules control outbound traffic from VMs.

Ephemeral external IP: A public IP address automatically assigned to a VM that is released when the VM is stopped; not suitable for production use cases requiring a stable address.

Ephemeral internal IP: A private IP address automatically assigned from the subnet range each time a VM starts; not suitable when a stable internal address is required.

Firewall rule: A VPC-level rule that allows or denies traffic to or from VMs based on protocol, port, source, and destination, applied by priority number.

GKE (Google Kubernetes Engine): Google Cloud's managed Kubernetes service; subnets used for GKE require secondary IP ranges for pod and service addresses.

HA VPN (High Availability VPN): A Cloud VPN configuration with two gateway interfaces that provides a 99.99% uptime SLA when paired with a redundant peer gateway and two VPN tunnels.

IAM (Identity and Access Management): Google Cloud's system for granting permissions to principals at the project, folder, or organization level; referenced in firewall rules through target service accounts.

Ingress: Network traffic that enters a VPC network from an external source; firewall ingress rules control inbound traffic to VMs.

IPsec (Internet Protocol Security): The tunneling and encryption protocol used by Cloud VPN to secure traffic traveling over the public internet between two endpoints.

MX record: A DNS resource record type that specifies the mail server(s) responsible for accepting email for a domain, along with a priority value.

NAT (Network Address Translation): The process of rewriting the source IP address of outbound packets from a private IP to a public IP; Cloud NAT performs this at the network layer without a proxy.

NS record: A DNS resource record type that delegates authority for a zone to specific name servers.

Partner Interconnect: A Cloud Interconnect option that provides connectivity to Google's network through a supported service provider, with bandwidths from 50 Mbps to 50 Gbps.

Point of Presence (PoP): A physical facility where Google's network meets the broader internet and where Dedicated Interconnect connections are terminated.

Port allocation: A Cloud NAT setting that controls the minimum and maximum number of source ports available per VM for outbound connections, affecting the total number of simultaneous connections.

Private DNS zone: A Cloud DNS managed zone visible only to specified VPC networks, used for resolving internal hostnames that should not be exposed on the public internet.

Private Google Access: A subnet-level setting that allows VMs without external IP addresses to reach Google APIs and services (e.g., Cloud Storage, BigQuery) over Google's internal network.

PTR record: A DNS resource record type used for reverse DNS lookups, mapping an IP address back to a hostname.

Public DNS zone: A Cloud DNS managed zone that is accessible from the public internet, used for external-facing services.

RFC 1918: The IETF standard defining the private IP address ranges (`10.0.0.0/8` , `172.16.0.0/12` , `192.168.0.0/16`) used inside VPCs and subnets.

Secondary IP range: An additional IP address range configured on a GKE subnet to provide addresses for Kubernetes pods and services, separate from the primary subnet range used by VMs.

SLA (Service Level Agreement): A contractual commitment to a minimum level of service availability; Cloud DNS offers 100% SLA for authoritative serving, and HA VPN offers 99.99%.

SOA record: The Start of Authority DNS record that contains metadata about a DNS zone, including the primary name server, contact email, and timing parameters.

SPF (Sender Policy Framework): An email authentication mechanism published as a TXT DNS record that specifies which mail servers are authorized to send email for a domain.

SRV record: A DNS resource record type that specifies the location (host and port) of a service, used for service discovery.

Static external IP: A public IP address reserved independently of any VM that persists until explicitly released; billed even when not attached to a resource.

Static internal IP: A private IP address from a subnet range explicitly reserved for a specific resource such as a database server or internal DNS server, ensuring it does not change.

Subnet: A regional IP address range within a VPC network to which VM instances are attached; subnets can only be expanded (not shrunk) after creation.

Tag spoofing: The act of a malicious VM applying a network tag it should not have in order to match permissive firewall rules; mitigated by using target service accounts in firewall rules instead of tags.

Transitive routing: The ability of network A to reach network C through network B; VPC peering does not support transitive routing — each VPC pair must create its own peering connection.

TTL (Time to Live): The number of seconds a DNS resolver should cache a record before re-querying the authoritative server; lower values allow faster propagation of changes.

TXT record: A DNS resource record type that holds arbitrary text, commonly used for domain verification, SPF, and DKIM records.

VPC (Virtual Private Cloud): Google Cloud's isolated virtual network that provides private IP space, routing, firewall rules, and connectivity services for cloud resources.

VPC Flow Logs: A feature that captures a sample of network traffic flowing through a subnet's VM interfaces and sends it to Cloud Logging, used for network analysis and security monitoring.

VPC peering: A networking connection between two VPC networks (in the same or different projects or organizations) that enables private IP communication; requires both sides to create a peering connection and does not support transitive routing.